Anh T.M Truong

+1 (352) 756 0006 • truongtranmaianh@gmail.com
maicarem.github.io • linkedin/maianhtruong1509 • scholar.google.com/maianhtruong1509

Education

University of Florida • Florida, USA — May 2027
Ph.D. Student of Operations Research, ISE
- Research Focus: Interdiction Game, Bilevel Optimization, Integer Programming.
- Related Coursework: Smart Grids For Sustainable Energy, Mathematical Programming, Machine Learning, Stochastic Process, Stochastic Optimization, High-Dimensional Data Analytics, Algorithms

Paris Sciences et Lettres University • Paris, France — September 2023
Master of Computer Science
- Thesis: Ring Star Problem under the failure of two hubs.
- Received combined fully-funded "PSL Fellowship" and "Excellence Scholarship" by French Embassy.

Vietnam National University • Ho Chi Minh city, Vietnam — August 2022
Bachelor of Industrial and Systems Engineering
- Valedictorian (Class of 2022) (1/100), fully-funded scholarship, Ranked **first** in National Entrance Examination.
- Received Vietnam Outstanding Female Student in Science and Technology (only 20 female students nationwide selected, featured in press).

Skills

- **Programming language** Julia, Python • **Tools** SQL, Matplotlib, Pandas, scikit-learn, Airflow, LaTeX
- **Mathematical programming tools** JumP, Gurobi, CPLEX, AMPL

Research Experience

Reasearch Assistant – Center for Applied Optimization – University of Florida — September 2023 - Present
Advisor: Dr. Jorge. A. Sefair
- **Research topic 1:** Service Agent Transport Problem. Formulated a flow-based model for scheduling with multiple transport and service agents.
- Introduced branch-and-cut approach with critical path cuts. Proposed acceleration techniques (dynamic programming, greedy algorithm) to transform an approximate solution into a feasible one.
- Achieved 40% performance improvement and 60% time reduction over state-of-the-art.
- **Research topic 2:** Generalized Network Interdiction Min-Cost Flow Problem. Utilized the k-shortest path to predict the interdictor's behavior on solving interdiction.

Research Intern – Paris-Dauphine, PSL — February 2023 - September 2023
Advisor: Dr. André Rossi & Dr. Sonia Toubaline
- **Research topic:** Branch-and-Benders-Cut on network optimization subject to uncertainty. Proposed integer programming models for survivable network design, implemented by Julia and Gurobi.
- Devised and implemented accelerating techniques for the Branch-and-Benders-Cut approach, resulting in a faster convergence rate by 17%.

Reasearch Assistant – Vietnam National University — September 2020 - September 2022
Advisor: Dr. Hop Nguyen Van
- **Research topic:** Metaheuristics for supply chain problems (assembly line balancing, port scheduling problem, job shop scheduling).
- Implemented genetic algorithm and particle swarm optimization for research projects using Python, outperforming existing state-of-the-art solutions.

Publications

Journal
- A. T. M. Truong, N.V. Hop, "Matheuristics for mixed-model assembly line balancing problem with fuzzy stochastic processing time", **Applied Soft Computing, IF= 8.7**
- D. C. Hop, N. V. Hop, and A. T. M. Truong, "Adaptive particle swarm optimization for integrated quay crane and yard truck scheduling problem", **Computers & Industrial Engineering, IF= 7.18**,
- A. T. M. Truong, Sonia Toubaline and André Rossi, "Ring Star Problem under the failure of two hubs" (Submitted to **Computers and Operations Research**, Revise and Resubmit).
- A. T. M. Truong, Jorge A. Sefair, "Service Agent Transport Problem" (In preparation).

Conference
- A. T. M. Truong, Sonia Toubaline and André Rossi, "Ring Star Problem under the failure of two hubs", ROADEF 2024, Amiens, France.

Work Experience

Demand Planner – Yes4All — February 2022 - August 2022
- Optimized SQL queries and visualized metrics using Apache Superset, enhancing data operations by 25%.
- Analyzed sales data with Python and SQL, reducing excess sporting goods inventory costs by 15%.

Engineering Intern – Intel — August 2020 - August 2021
- Implemented VBA automation to manage inventory records, increasing accuracy and efficiency.
- Developed a PowerApps application to streamline headcount reporting at the factory.